A Reproducible Toy Study of Multi-Target Manuscript Generation

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Scientific manuscripts are often copied into separate directory trees when only repository branding differs. That duplication makes later corrections difficult to propagate and encourages source drift. We present a toy manuscript that uses one LaTeX class, one bibliography style, and one content source to create bioRxiv-, arXiv-, Zenodo-, and neutral-profile PDFs. The profiles change only the footer label; manuscript content, typography, authorship, citations, and figure sources remain identical. This small example is intended as a development fixture and as a starting point for a real manuscript, not as evidence of compliance with every repository's current submission policy.

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Introduction

Reproducible research depends on preserving both the scientific record and the process that produced it. Shared conventions for data, metadata, and software improve the ability of other researchers to inspect and reuse scholarly outputs (1). Manuscript production deserves the same care. When an article is copied once for each intended repository, prose and references may be corrected in one copy but not another. The resulting files look related while quietly representing different states of the work.

The template tested here treats the repository name as configuration rather than content. A target profile is selected at build time, while the manuscript is read from the same `input/Manuscript.tex` file. The target may also be selected with a native class option for authors who compile interactively. This separation keeps repository presentation at the edge of the system and keeps scientific content in one authoritative source.

Template design

The design has three components. First, the manuscript contains all authors, affiliations, narrative text, captions, and citations. Second, the `MultiTargetManuscript` class contains shared typography and four small target profiles. Third, the build helper passes the requested target to LaTeX before the class is loaded. The bibliography and source-native figure are shared by every build (Fig. 1).

The four targets are arXiv, bioRxiv, Zenodo, and neutral. The first three values produce a visible footer label. The neutral value removes the label entirely. Existing manual controls are retained: `\journalname{...}` supplies a custom label, and `\nojournalname` suppresses it. These controls make the template usable for journal drafts without creating another class file.

Validation strategy

Document automation should be tested at more than one level. A successful LaTeX exit status establishes that the source can be processed, but it does not show that the correct profile was selected or that the footer is legible. The example therefore supports four complementary checks:

1. compile every supported target from a clean auxiliary directory;
2. inspect logs for undefined citations, references, and fatal errors;
3. extract text from each PDF and confirm the expected label behavior; and
4. render the PDF pages and inspect typography, spacing, and clipping.

This layered approach follows the broader principle that computational outputs should be accessible, inspectable, and reproducible (2). It also distinguishes a template fixture from a repository submission test. The fixture proves that target selection changes the local output as designed. It does not prove that an external service will accept a future upload, because external requirements and supported TeX environments can change.

Illustrative result

All four outputs are expected to have identical pagination and body text. The arXiv build displays a stylized arXiv label; the bioRxiv build preserves the stylized bioRxiv mark from the source template; and the Zenodo build displays a plain Zenodo label. The neutral build retains the author, date, short title, and page information but contains no repository name. Conditional separator logic prevents a doubled bar or unexplained gap when the label is absent.

Because the target selection occurs outside the manuscript content, a correction to a paragraph, reference, author list, or figure is automatically present in the next build of every profile. This is the principal maintenance advantage of the consolidated layout. The source tree is also small enough to package for an archive or transfer to another LaTeX environment without determining which of several manuscript copies is authoritative.

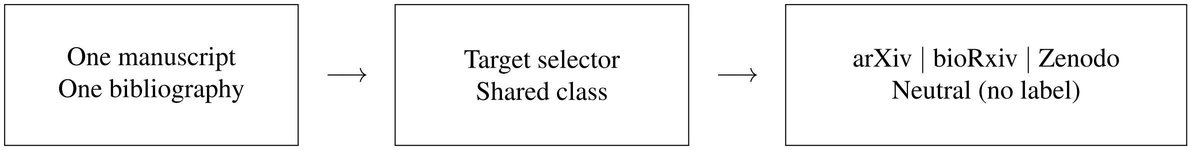

Fig. 1. One source tree generates four presentation profiles. The target selector changes only the repository label in the footer. The neutral profile suppresses that field and its adjacent separator, so it does not leave an empty visual artifact.

Limitations and use

Repository labels are presentational metadata. They do not imply endorsement by arXiv, bioRxiv, or Zenodo, and they do not replace each service's submission form or metadata requirements. Zenodo is a general-purpose repository rather than a journal with a mandatory article class, so its profile is deliberately a label on the shared layout. Authors should reserve or add a DOI separately when their deposit workflow calls for one.

For a real project, replace the toy prose, authorship, bibliography, and figure while retaining the directory layout and build interface. Authors can add ordinary LaTeX packages in the manuscript preamble. Changes to page geometry, title construction, bibliography formatting, or footer policy belong in the class and should be documented in its dated revision notes.

Conclusion

A target selector is sufficient to replace three nearly identical manuscript trees with one maintainable source. The resulting repository exposes the choice clearly, keeps supporting files synchronized, and allows neutral or custom output without another fork of the class. The example is intentionally small so that future template changes can be compiled and visually checked quickly.

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This manuscript is a fictional development fixture. Names, institutions, and email addresses are illustrative and do not describe real contributors.

Bibliography

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